Performance Testing

Date	27 September 2026
Team ID	SWTID-2026-2630
Project Name	AI Fit Track API
Maximum Marks	4 Marks

Performance Testing – Endpoint Response Benchmarks

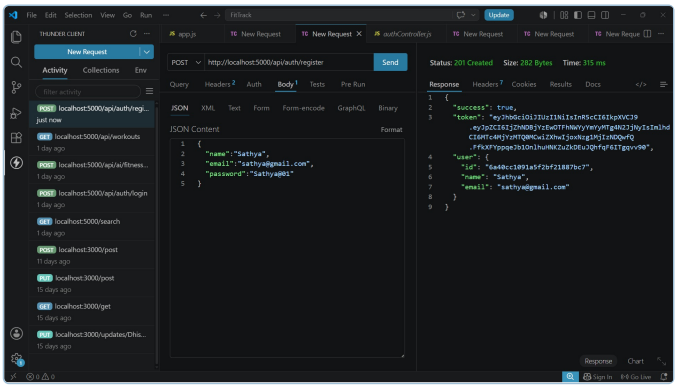
All 10 implemented endpoints were exercised with Thunder Client against the local MongoDB instance (`mongodb://localhost:27017/aifittrack`) on port 5000. Response time and payload size were recorded directly from the client for each call.

S.No	Endpoint	Method	Status	Response Time	Response Size
1	/api/auth/register	POST	201 Created	315 ms	282 B
2	/api/auth/login	POST	200 OK	189 ms	282 B
3	/api/auth/profile	GET	200 OK	75 ms	100 B
4	/api/workouts/ (add)	POST	201 Created	29 ms	303 B
5	/api/workouts/ (get all)	GET	200 OK	134 ms	315 B
6	/api/workouts/:id (get by id)	GET	200 OK	12 ms	303 B
7	/api/workouts/:id (update)	PUT	200 OK	94 ms	303 B
8	/api/workouts/:id (delete)	DELETE	200 OK	69 ms	57 B
9	/api/ai/workout-recommendation	POST	200 OK	**7.28 s**	1000 B
10	/api/ai/fitness-insights	POST	200 OK	**5.01 s**	275 B

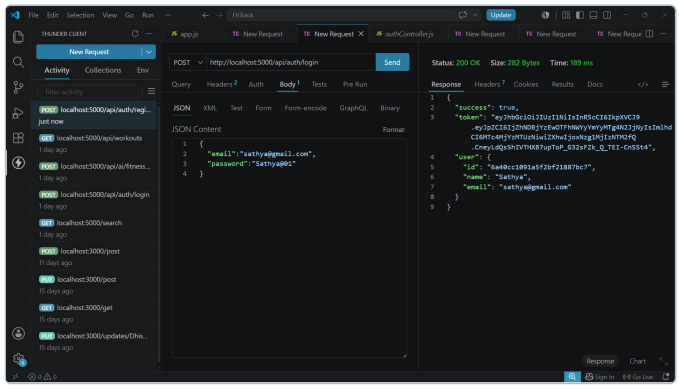
Observations

- All 8 MongoDB-only endpoints (auth + workout CRUD/search) return in well under 350 ms, consistent with a local database and lightweight Mongoose queries with no heavy aggregation.
- The two AI endpoints are an order of magnitude slower (5–7.3 seconds) because each call blocks on a live round trip to the external Google Gemini API for text generation – this is external network/model latency, not a bottleneck in the Express/MongoDB layer.
- `GET /api/workouts/:id` is the fastest call (12 ms), as expected for a single indexed document lookup by `_id` .
- Recommendation: for a production deployment, the two AI endpoints should show a loading state client-side and could be moved to an async/queued pattern if request volume grows, since their latency is dictated by the Gemini API rather than by AI Fit Track's own code.

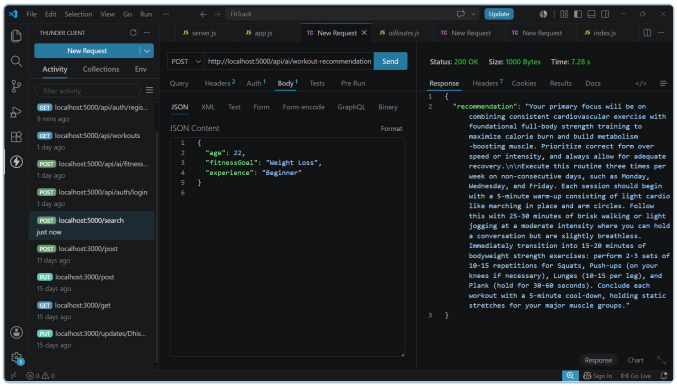
Evidence – Sample Test Executions



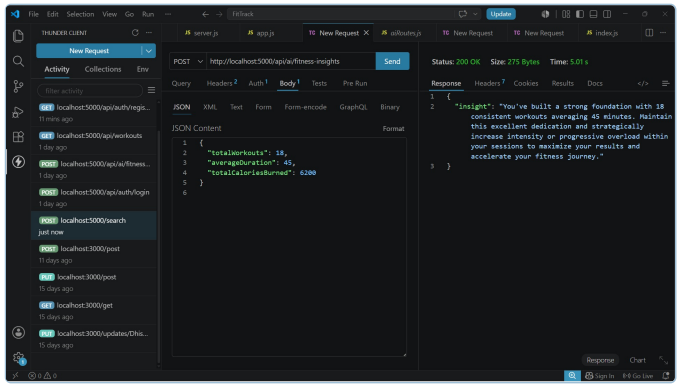
Register – 201 Created, 315 ms



Login – 200 OK, 189 ms



AI Workout Recommendation – 200 OK, 7.28 s



AI Fitness Insights – 200 OK, 5.01 s